BitBot XL and Classic cheat sheet

Library: **bitbotXL.py** or **bitbotClassic.py** — upload it with your **main.py**



Getting started

```
from microbit import *
import bitbotXL as bot          # BitBot XL
# import bitbotClassic as bot   # BitBot Classic

bot.move_forward(50, 1000)
print(bot.get_distance())
```

Import the library that matches your robot. Always use the `bot.` prefix.

Movement

Speed is **0–100** (default 50). `duration_ms` = how long, then it stops. Leave it out and the robot keeps going until `stop()`.

`move_forward(speed, duration_ms)`	Drive forwards
`move_backwards(speed, duration_ms)`	Drive backwards
`turn_left(speed, duration_ms)`	Spin anticlockwise
`turn_right(speed, duration_ms)`	Spin clockwise
`stop()`	Stop both motors

```
bot.turn_left(50, 600)   # spin 0.6 s
bot.move_forward()       # keep going...
sleep(1000)
bot.stop()
```

LEDs

12 RGB LEDs numbered **0–11**: 0–5 left arm, 6–11 right arm. `led` = one LED, `leds` = all of them.

`led_on(led, colour)`	Turn on one LED
`led_off(led)`	Turn off one LED
`leds_on(colour)`	Turn on all LEDs
`leds_off()`	Turn off all LEDs

Colours: `(r, g, b)` each 0–255 (keep under ~60, they're bright), or `bot.RED` `GREEN` `BLUE` `YELLOW` `PURPLE` `WHITE`.

```
bot.leds_on(bot.RED)
bot.led_on(3, (0, 0, 50))
```

Buzzer

`buzzer(duration_ms)`	Beep (fixed pitch, default 200 ms). Waits until the beep ends.

Sensors

`get_distance()`	Distance in **cm**, or **-1** if no echo (check for this!)
`get_left_light()` `get_right_light()`	Light level **0** (dark) to **1023** (bright)
`get_left_line()` `get_right_line()`	**1** = over the line, **0** = not

Ultrasonic works best on hard, flat objects. Leave ~50 ms between readings. Light values depend on the room — measure your own thresholds.

Example: stop before a wall

```
bot.move_forward(40)
while True:
    d = bot.get_distance()
    if 0 <= d < 15:
        bot.stop()
        bot.buzzer(300)
        break
    sleep(60)
```

XL or Classic?

Both libraries have exactly the same functions, so only the `import` line changes between robots.

BitBot XL	Upload `bitbotXL.py`
BitBot Classic	Upload `bitbotClassic.py`

Classic only: the line sensors share their pins with the buttons (left sensor with B, right sensor with A), so don't use the buttons in a program that follows a line.